

Stat 244: Statistical Theory and Methods 1 — Autumn 2026 syllabus

Course description This course is the first quarter of a two-quarter systematic introduction to the principles and techniques of statistics, as well as to practical considerations in the analysis of data, with emphasis on the analysis of experimental data. This course covers tools from probability and the elements of statistical theory. Topics include the definitions of probability and random variables, binomial and other discrete probability distributions, normal and other continuous probability distributions, joint probability distributions and the transformation of random variables, principles of inference (including Bayesian inference), maximum likelihood estimation, hypothesis testing and confidence intervals, likelihood ratio tests, multinomial distributions, and chi-square tests. Examples are drawn from the social, physical, and biological sciences.

See <https://www.stat.uchicago.edu/~yibi/IntroStat/> for help deciding between Stat 200, 220, 234, 244, and 24410.

Course info

- Instructor: Rina Barber, rina@uchicago.edu
- TAs: Yijun Fan, yjfan@uchicago.edu
Shuo Zhang, shuozhang2002@uchicago.edu
Laura DeFalco, lauradefalco@uchicago.edu
Tannistha Mondal, tannistha@uchicago.edu
- The main course page is on Canvas and you can find all lecture videos, slides, assignments, etc there. Homework will be handed in and graded on Gradescope. We will also use Ed Discussion for Q&A.
- Class will be held in-person. Office hours will be in-person or on Zoom (as listed below).
- Class time & location: Mon/Wed 1:30-2:50, 1155 Building (Social Sciences Research Center) room 142
- Weekly office hours schedule:
 - Mondays 5-6pm (TAs — Jones 226)
 - Tuesdays 5-6pm (TAs — Jones 226)
 - Tuesdays 6-7pm (TAs — zoom)
 - Fridays 11am-12pm (Rina — Jones 214 — except that on Week 1, will instead be 9-10am)
 - Problem session, Fridays 3-4pm (TAs — Jones 226)

Exams

- The midterm exam is **tentatively** scheduled for **Thursday Oct 29, 6:00pm** (we are waiting to receive a room assignment so the date/time is still uncertain). Please contact the instructor immediately if you may need an alternative exam time.
- The final exam will be given during the registrar-assigned day & time (which the registrar will announce later on). Please contact your instructor immediately if you may need an alternative exam time.

Handing in assignments

- Assignments are due on Wednesdays at 1:30pm (at the start of class).
- At the end of the quarter, the lowest problem set grade (or one missing grade) will be dropped. We cannot excuse any missed problem sets beyond the one that is dropped.
- Late assignments will be accepted with a penalty of 4% per hour (late time is rounded up, i.e., one minute late counts as one hour late). We cannot offer any additional extensions.
- Assignments are submitted and graded via Gradescope (which can be accessed from the Canvas course page). Students are required to tag pages for each problem.
- If you are having trouble uploading to the website and run out of time, please email your work to the instructors or TAs before the time problem set is due as proof of completion. The time of your email will count as the time of your problem set submission. We do not accept the time stamp of the file on your computer as proof of completion.

Contacting us We will aim to reply to all questions within 24 hours.

- For any questions about the material or homework or exams (aside from regrade requests), please contact us through the discussion boards on Ed Discussion.
 - Please post publicly if possible (e.g., questions about material, clarification on problems sets, questions to help understand a midterm problem after the exam has been graded, etc). Note that you can choose to post anonymously but your name will still be visible to the instructor/TAs.
 - Alternatively, you can write a private post, visible only to the instructor/TAs (e.g., if you help on a homework problem but posting your question would reveal too much of your work).
- For any questions about your graded homework or exams, please submit a regrade request on Gradescope.
- For other questions such as enrollment, prerequisites, accommodations, makeup times for exams, etc, please contact the instructor by email.

Grading The final grade will be determined by the following approximate weights (exact weights will be determined later in the quarter):

- Problem sets: 25% (with lowest grade, or one missing assignment, dropped)
- Midterm exam: 35%
- Final: 40%
- If the final exam grade is higher than the midterm grade, it will instead be given more weight (30% midterm, 45% final)

Textbook & resources The textbook for this course is:

- *Mathematical Statistics and Data Analysis*, Rice, 3rd edition.

The textbook can be rented from the publisher:

<https://www.cengage.com/c/mathematical-statistics-and-data-analysis-3e-rice/9780534399429PF/>

We will not assign homework problems from the textbook, but the book is a very useful reference for the material, and we will also be posting suggested practice problems from the textbook.

Collaboration guidelines & plagiarism policy For problem sets, students are free to discuss the problems and collaborate on strategies for solving the problems, but all writing, code, etc, should be done completely on your own. For example, working out a solution as a group, then transferring it to your own page, is not acceptable. Similarly, any use of AI tools should follow the same guidelines: students may use such tools to answer questions or help develop initial ideas, etc, but handing in solutions that are LLM-generated (even if reviewed, edited, or paraphrased by the student) is not permitted. No collaboration or discussion of any kind, or AI use of any kind, is allowed on the exams.

Any copied material (from websites, published materials, LLM output, or another students' work) that is handed in without attribution will be considered to be plagiarism and will be reported to the appropriate university department. Feel free to reach out to the instructor or TAs if you have any questions about what is appropriate for collaboration or online resource use.

Please consult the student manual on university policies and regulations that make it clear that the University will not tolerate cheating and plagiarism: <https://studentmanual.uchicago.edu>

Special Accommodations The University of Chicago is committed to ensuring equitable access to our academic programs and services. Students with disabilities who have been approved for the use of academic accommodations by Student Disability Services (SDS) and need a reasonable accommodation(s) to participate fully in this course should follow the procedures established by SDS for using accommodations. Timely notifications are required in order to ensure that your accommodations can be implemented. Please contact the instructor to discuss your access needs in this class after you have completed the SDS procedures for requesting accommodations.

Phone: (773) 702-6000, Email: disabilities@uchicago.edu